

ENV 6128

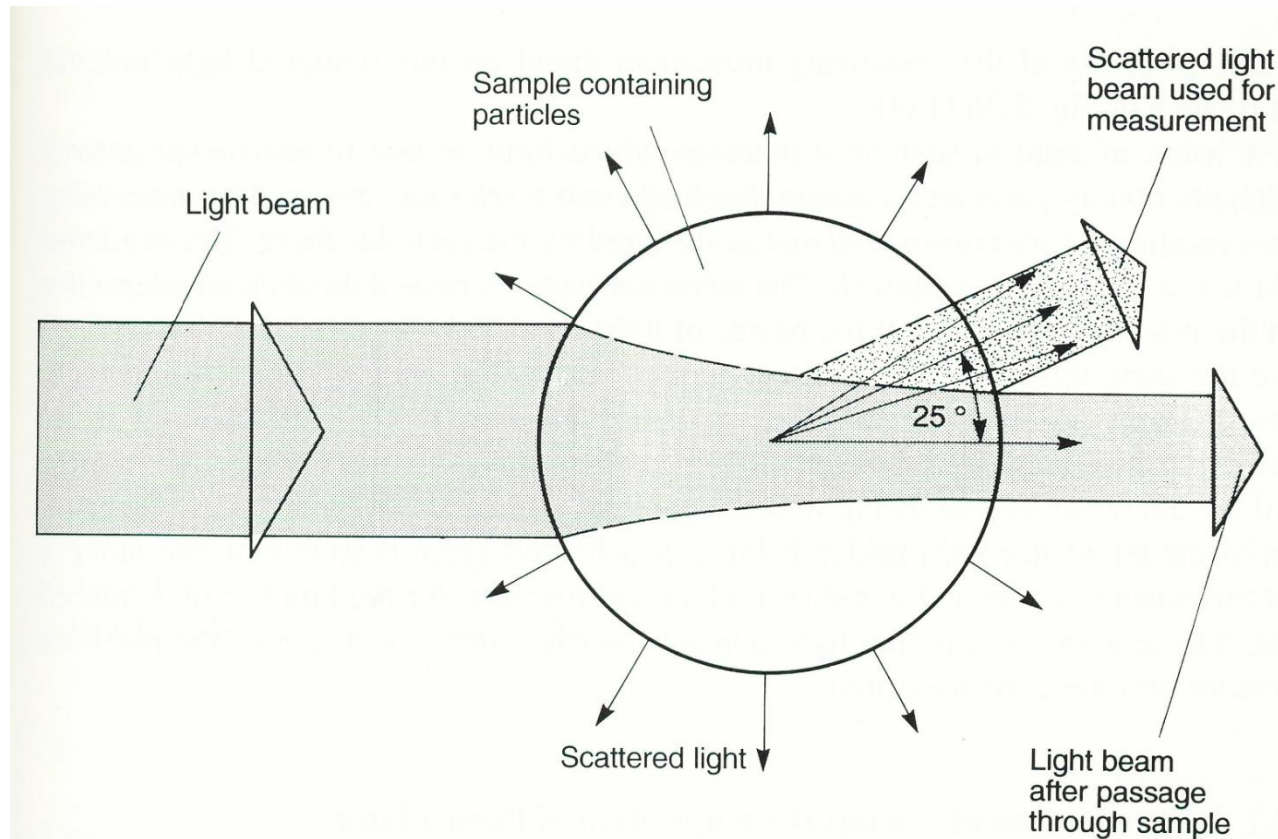
Smart Air Quality Monitoring & Air Pollution Control

Today's Topics

- Measurement technologies for PM mass
- Expectation
 - Understand different technologies for measuring PM mass
 - Understand the strengths and weakness of these technologies

Measuring particles - light scattering and absorption

- Particle optical properties depend on particle size, shape, and composition
- Absorption and scattering are wavelength dependent



Measuring particles - light scattering and absorption

- Nephelometers measure light scattering
- Aethalometers measure light absorption
 - Calibration using a reference aerosol
 - Mostly for black carbon
- Optical particle counter also measures light scattering



Magee Scientific AE33 aethalometer



Ambilabs 2-WIN (Two Wavelength Integrating Nephelometer)

Measuring particles - light scattering

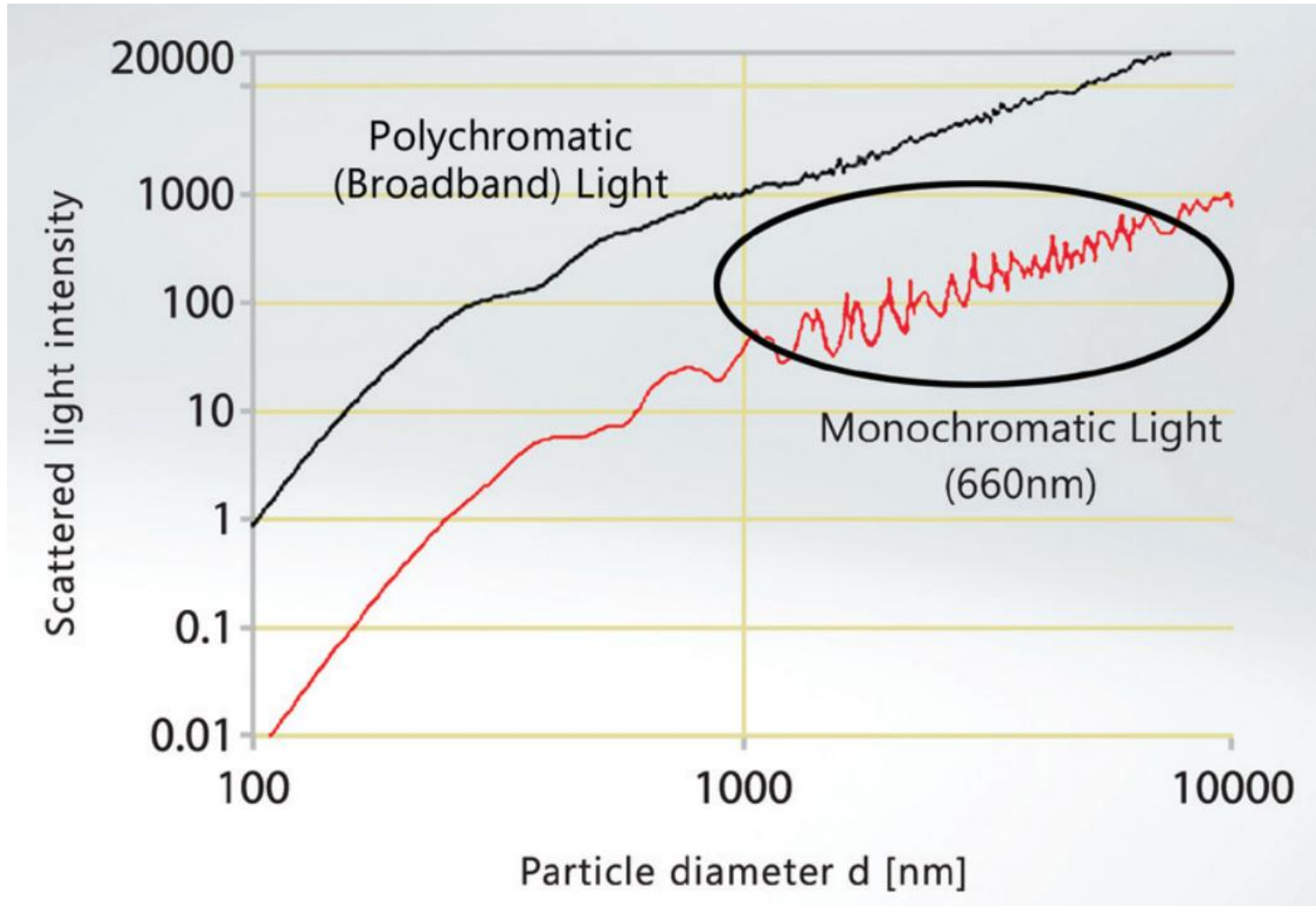


GRIMM EDM 180
0.25 - 32 μm
31 size channels
Single wavelength (660 nm)

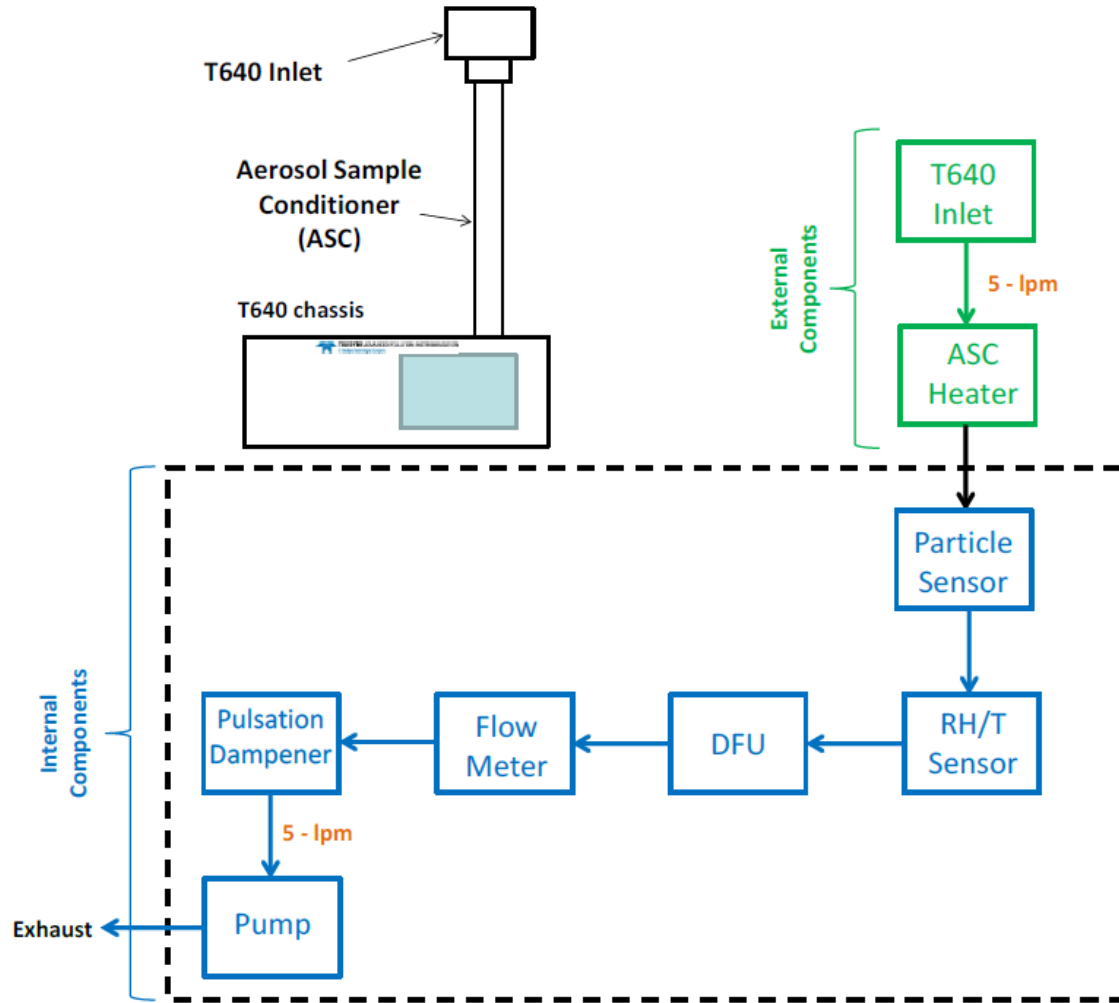


Teledyne API T640
0.18 – 20 μm
256 size channels

Teledyne T640



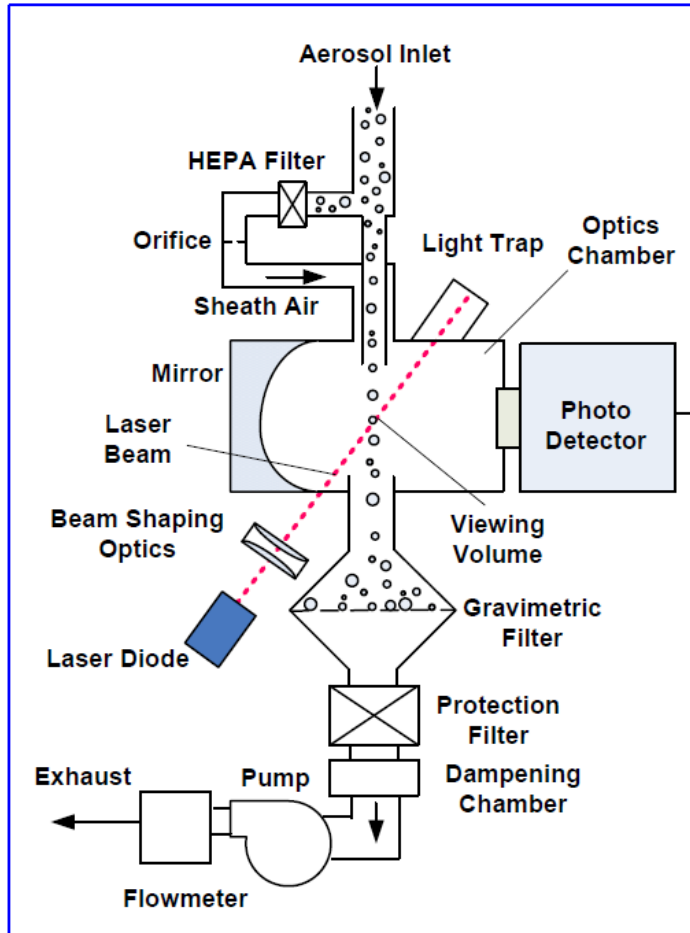
T640 Pneumatics



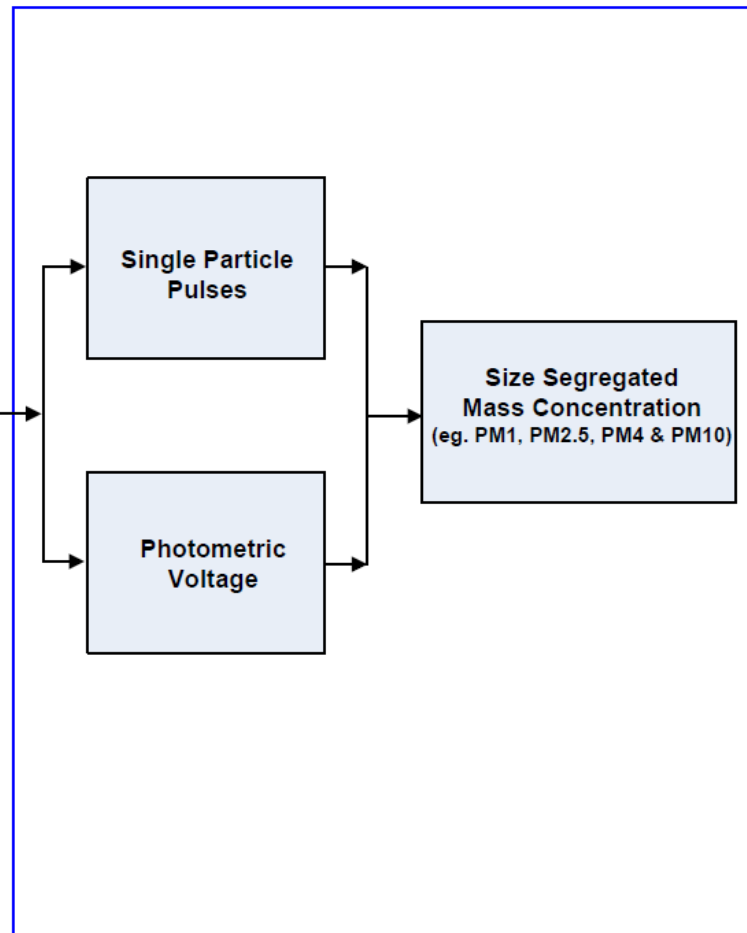
T640 Manual Fig 2-22
DFU: disposable filter unit

DustTrak DRX

Aerosol Measurement



Signal Acquisition and Processing



TSI application note expmn-002

Note: TSI DustTrak (only DRX) measures
both particle cloud and single particle

Cloud vs single particle



Single particle counting

Particle counter / spectrometer

Gives particle # and size

Not good for high conc



Cloud of particles

Photometer

Derive mass based on
calibration particle

Non-light based PM mass instrument

Measuring PM - Filtration

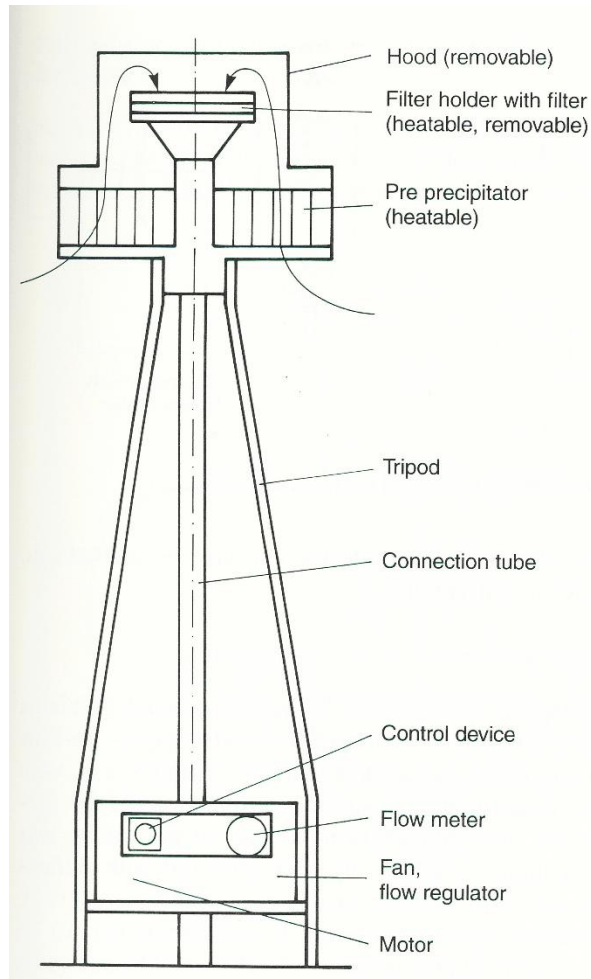
- Commonly used method for particle collection
 - AKA. Gravimetric method
- Forced air flow through a filter
- Mass collected on filter is weighed
 - concentration ($\mu\text{g}/\text{m}^3$) = mass collected [μg] / air volume (m^3)
- Also chemical analysis, microscopy for identification
- Filter materials depends on particle composition and particle size
- All FRM (Federal Reference Methods) PM instrument are filter based
 - FRM/FEM list: <https://www.epa.gov/amtic/air-monitoring-methods-criteria-pollutants>

Filter sample



High-volume particle sampler

- Extractive collection, typically sample for 24 hours



Tisch TE-6070 series

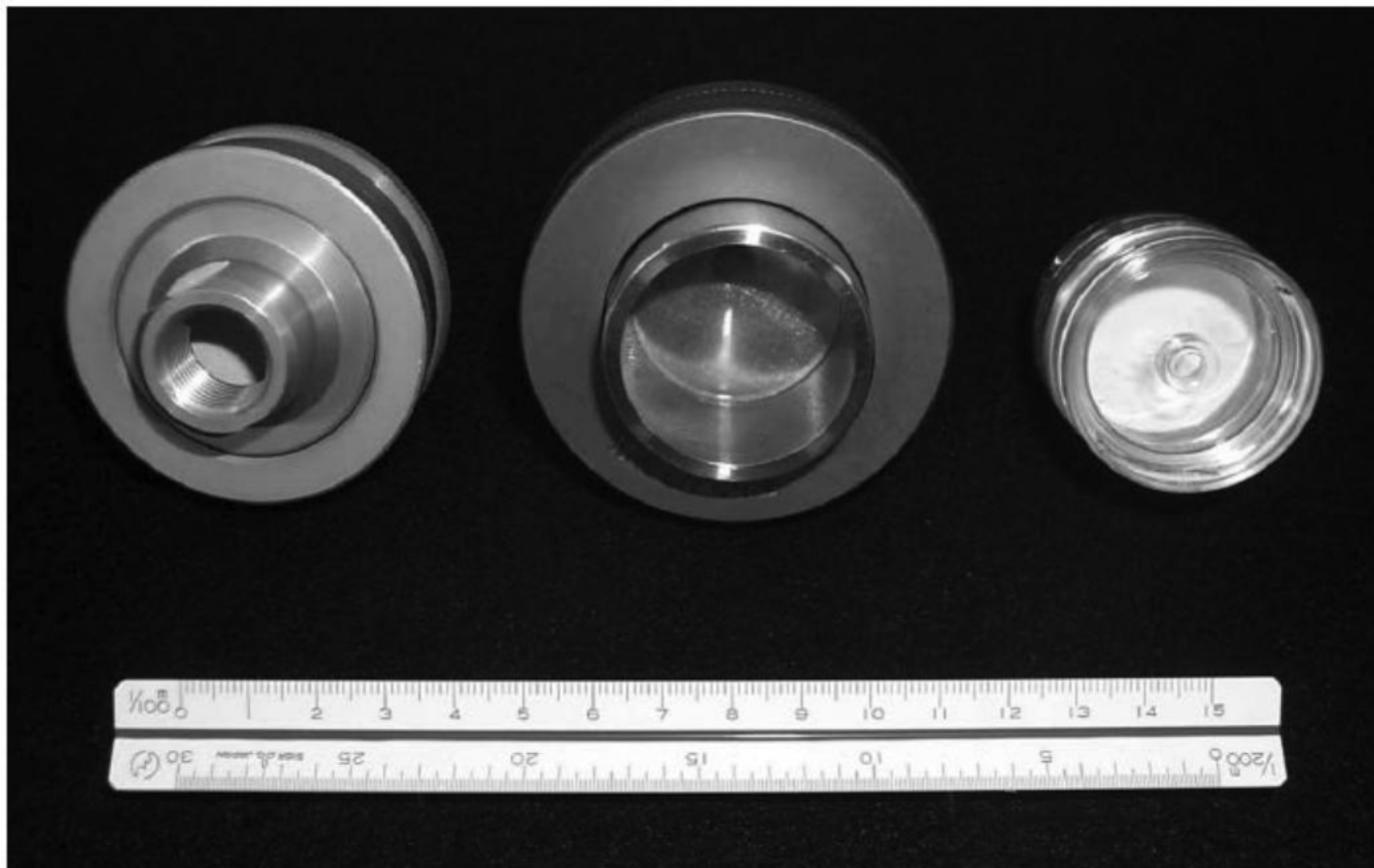


Figure 7-2 Examples of filter holders available commercially for aerosol collection (from left): in-line, **GEL**, 47 mm; open-face, **MIL**, 47 mm; and closed-face cassette, **GEL**, 25 mm (can also be used open-faced). Photograph courtesy of H. T. Kim, Gwangju Institute of Science and Technology.

Filter holder



GBI filter holders
Inlet cone



Pall filter holders



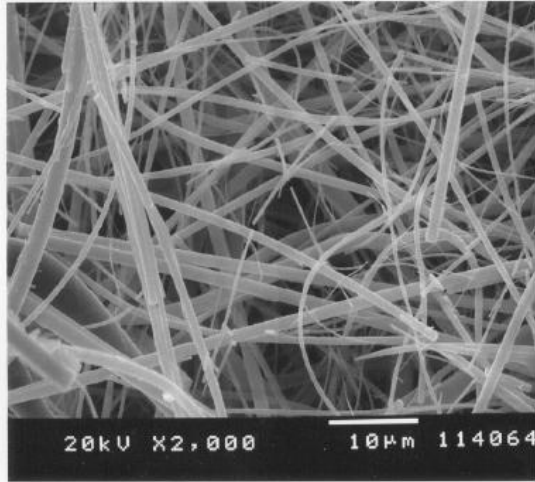
Most popular sizes: 12, 25, 37, 47, 100 mm diameter

Filter media

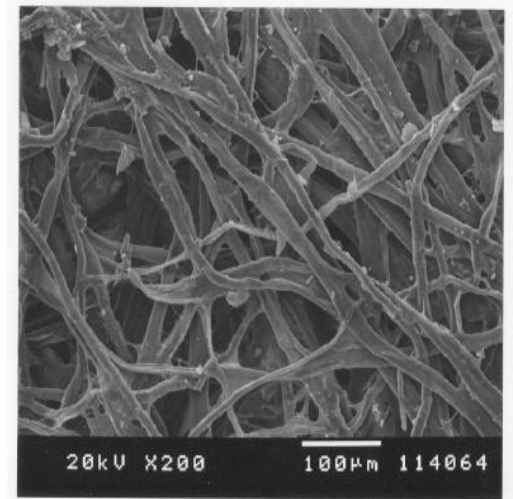
- Fibrous: glass, cellulosic, polymeric, metal, carbon, ceramic
- Membrane:
 - Polymeric: Teflon, cellulose ester, polysulfone, polyvinylidene, nylon, polycarbonate
 - Sintered metal: stainless steel, nickel, hastelloy, silver, titanium,
 - Ceramic: aluminum oxide, silicon carbide
 - Gelatin
- Fabric: cotton, fiberglass, polyester, composite synthetic, PTFE
- Foam: polymeric, metal
- Granular bed: sand

Filter media

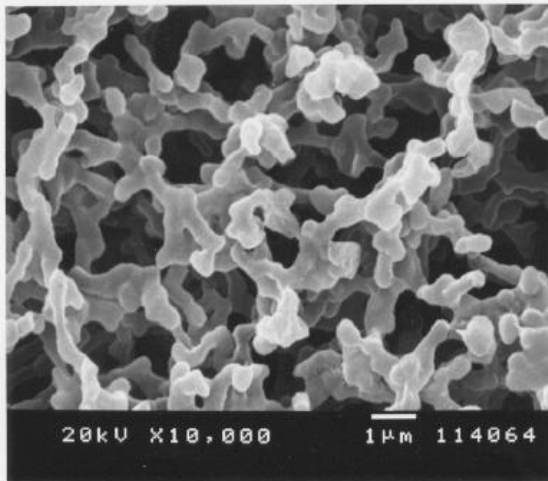
Glass
fiber



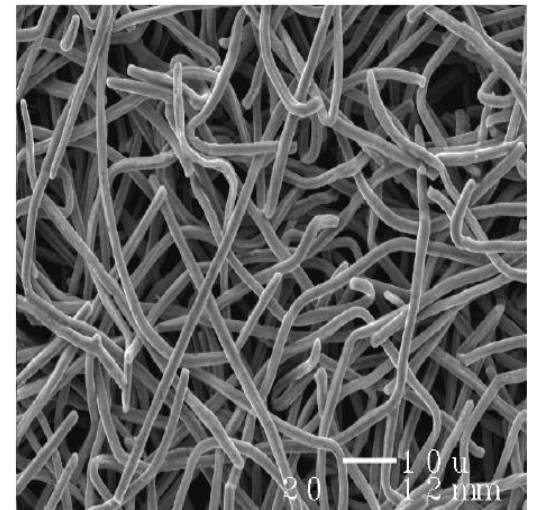
Cellulosic
fiber



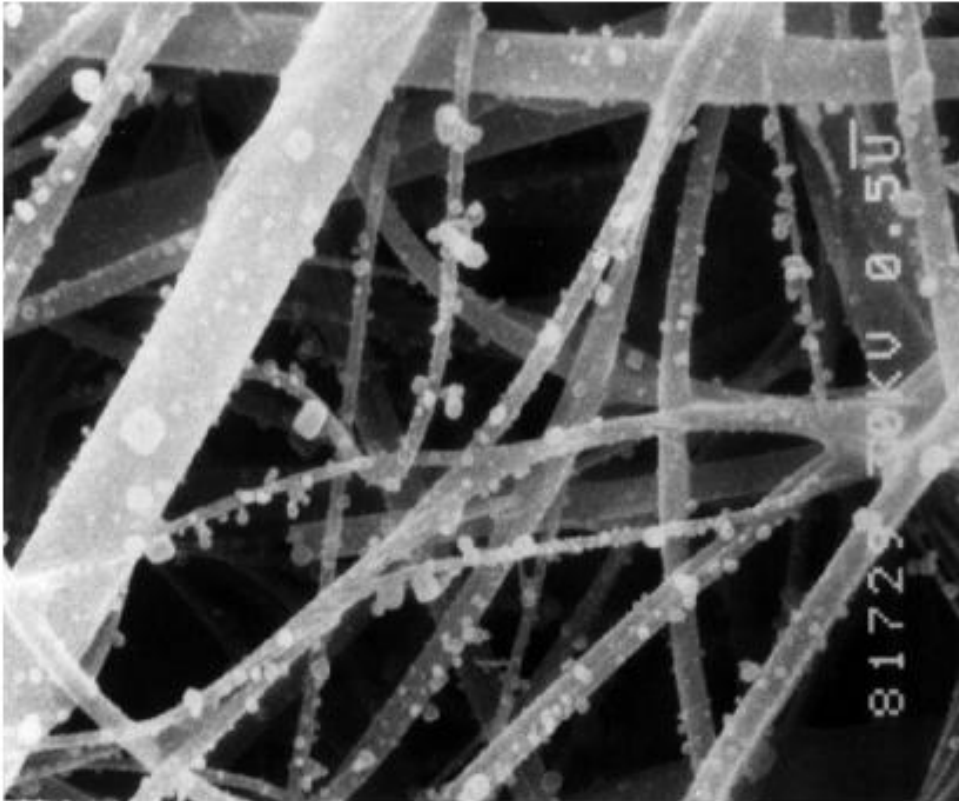
Cellulose
ester



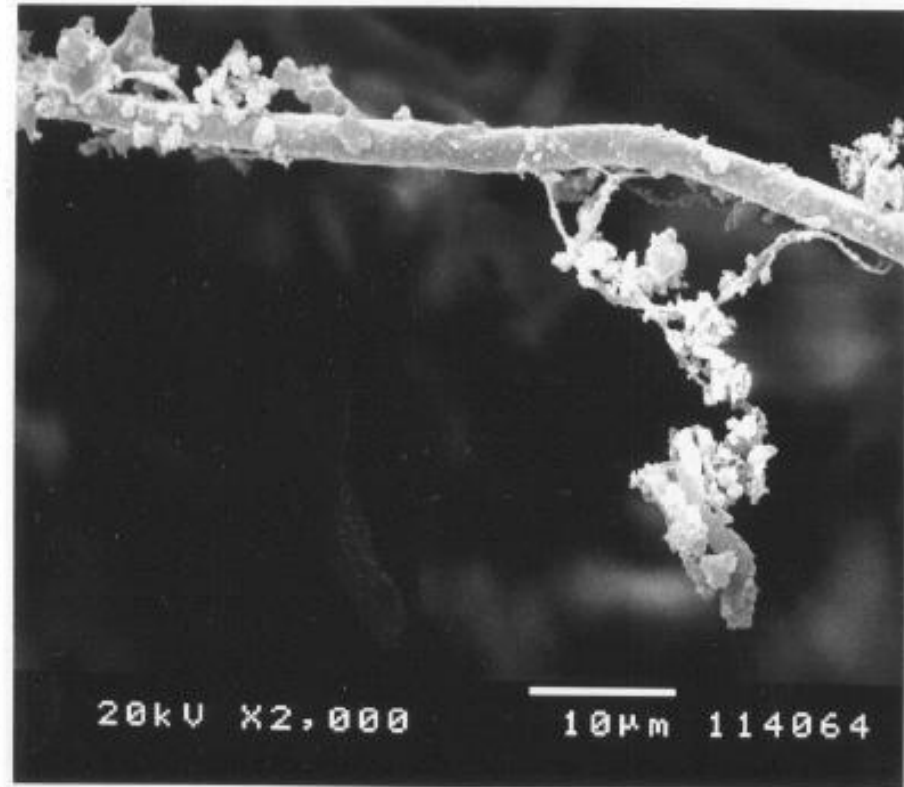
Metal
fiber



Particle laden fibrous filter

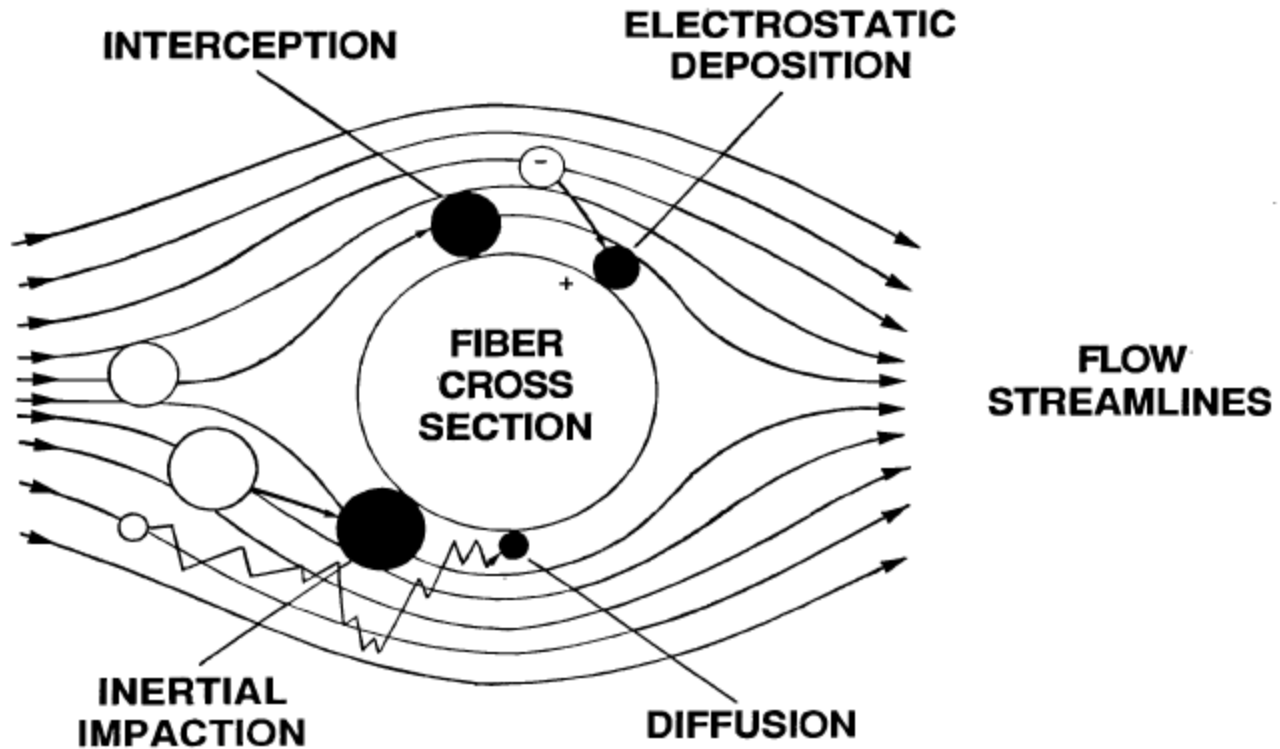


HEPA-glass fiber



Cotton synthetic fiber

Particle capture mechanism



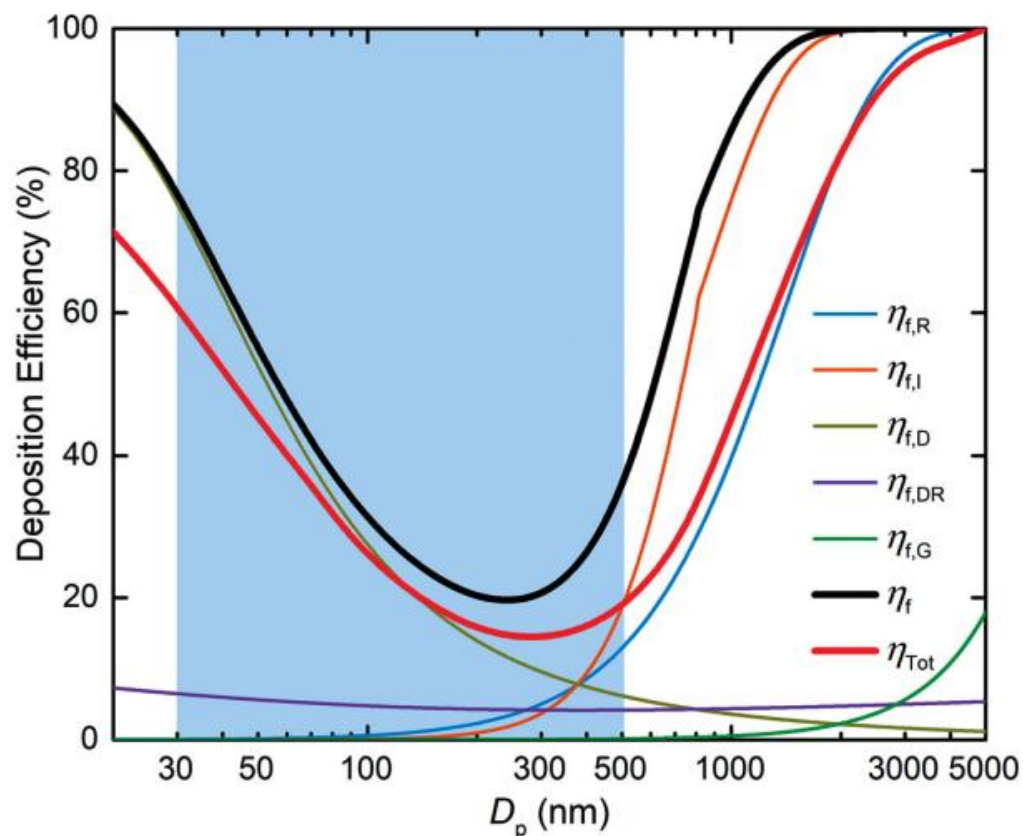


Figure 1. Particle deposition efficiencies due to different mechanisms and total efficiency in a filter with a fiber diameter ($D_f = 5 \mu\text{m}$), a thickness ($t = 1 \text{ mm}$), volume fraction of fibers ($\alpha = 0.05$), and a face velocity ($U_0 = 0.1 \text{ m/s}$), and a comparison against the particle deposition efficiencies in the human respiratory system. The shaded area shows the submicron size range of interest in this study. $\eta_{f,R}$, $\eta_{f,I}$, $\eta_{f,D}$, $\eta_{f,DR}$, $\eta_{f,G}$ stand for filter efficiency for interception, impaction, diffusion, diffusion-interception interaction, gravitational settling mechanisms, respectively. η_f stands for total filter efficiency, η_{Tot} stands for total particle deposition efficiencies in the human respiratory system.

Filter sampling artifacts

- Moisture effects

TABLE 7-5 Average (Avg^a), Standard Deviation (SD), and Coefficient of Variance (CV) of Mass Readings for Sampling Filters in Uncontrolled and Controlled Temperature and Humidity Environments

Filter Type	Uncontrolled Temperature and Humidity ^b	Controlled Temperature and Humidity ^b
Glass fiber filter	Avg = 87.933 mg SD = 0.0628 mg CV = 0.0714%	Avg = 87.844 mg SD = 0.0118 mg CV = 0.0134%
Teflon membrane filter	Avg = 244.371 mg SD = 0.0047 mg CV = 0.0019%	Avg = 244.370 mg SD = 0.0049 mg CV = 0.0020%
PVC membrane filter	Avg = 13.259 mg SD = 0.0021 mg CV = 0.0160%	Avg = 13.257 mg SD = 0.0023 mg CV = 0.0174%
MCE membrane filter	Avg = 41.582 mg SD = 0.1132 mg CV = 0.2722%	Avg = 41.435 mg SD = 0.0431 mg CV = 0.1040%

Source: Reprinted from Tsai et al. (2002) with permission.

^aThe same four filters of each type were weighed in the controlled and uncontrolled environments.

^bFilters were equilibrated for 24 h at 40%–45% RH prior to weighings.

Filter sampling artifacts

- Nonaqueous Vapor Adsorption Artifacts – In addition to water, other vapors can be adsorbed to filter elements or material previously collected by filters.
- Volatilization of Material Collected on Filters – After droplets or particles containing volatile or semi-volatile materials are captured by a filter, they have the potential to evaporate into the air passing through the filter.

TABLE 7-1 Summary of the Salient Characteristics of the Main Types of Filters Commonly Utilized for Aerosol Measurement

Filter Type	Characteristics
Fibrous filters	• Mat/weave of fibers with diameters of 0.1–100 μm; cellulose or wood (paper), glass, quartz, and polymer fiber filters available. • Porosities of 60%–99%, thicknesses of 0.15–0.5 mm. • Particle collection throughout the depth of the filter from interception, impaction, and diffusion onto fibers. • High particle collection efficiencies require low air velocities. • Pressure drops lowest among all filters under comparable conditions.
Membrane filters.	• Microporous membranes with tortuous pores throughout the structure. • Polymer, sintered metal, and ceramic microporous filters available. • Pore sizes (determined from liquid filtration) in the range 0.02–10 μm. • Porosities of <85%, and thicknesses of 0.05–0.2 mm. • Particle collection through attachment to microscopic structures within the filter. • High collection efficiencies, but highest pressure drop among all filters.
Capillary pore filters.	• Thin polycarbonate films (10 μm) with cylindrical pores perpendicular to film surface, with diameters in the range 0.1–8 μm. • Low porosities in the range of 5%–10%. • Particle collection through impaction and interception near the pores and diffusion to walls of pores. • Collection efficiencies intermediate between fibrous and microporous membrane. • Pressure drops significantly higher than fibrous filters and comparable to or higher than microporous membrane filters for equivalent collection efficiency.

TABLE 7-2 Commercial Sources and Typical Cost Ranges for Fibrous Sampling Filters

Filter Material(s)	Variety of Characteristics Available	Commercial Sources ^a	Typical Cost Range ^b
Cellulose fiber	• Fine, medium, coarse void sizes	<i>FSI, WHA, MSI, MFS</i>	Low
Borosilicate glass fiber	• With organic binder • Without organic binder • With Teflon coating • With cellulose content	<i>FSI, WHA</i> <i>MSA</i> <i>GEL</i> <i>MIL</i> <i>PAL</i> <i>MFS, MSI</i>	Low
Quartz fiber	• Pure • With small percentage borosilicate glass	<i>WHA, FSI</i> <i>MFS</i> <i>GEL</i>	Medium
Polymer fiber	• Polypropylene 0.45-, 0.8-, 1.2-μm uniform pore sizes	<i>MSI</i>	Low

TABLE 7-3 Commercial Sources and Typical Cost Ranges for Membrane Sampling Filters

Filter Material(s)	Variety of Characteristics Available	Commercial Sources ^a	Typical Cost Range ^b
Cellulose membrane	• Cellulose nitrate • Mixed ester • Cellulose acetate • 0.02–8 μm pore size	<i>MIL, FSI</i> <i>GEL</i> <i>MFS, MSI</i> <i>S&S, WHA</i>	Medium
Teflon/PTFE membrane	• Pure • Polypropylene reinforced 0.2–1.0 μm pore size	Same as above	High
Polyester/polycarbonate/ Polypropylene membranes	• 0.02–15 μm pore sizes	Same as above	Medium
Silver membrane	• Pure metallic silver 0.2–5 μm	<i>OSM, FSI</i> <i>MIL</i>	High
Nylon membrane	• Pure nylon • Laminated or impregnated with Polypropylene for support 0.1–20 μm pore size	<i>GEL, FSI</i> <i>MSI</i>	Medium
Polyvinyl chloride membrane	• Pure PVC • PVC with acrylonitrile 0.45-, 0.8-, and 5.0-μm pore sizes	<i>GEL, FSI</i>	Medium

TABLE 7-6 Compilation of Common Applications, Advantages, and Disadvantages for Various Types of Fibrous Sampling Filters

Filter Type	Typical Applications	Advantages	Limitations
Fibrous filters (general)	Air quality sampling	• Low pressure drop during high-volume sampling • Low cost • High particulate loading capacity	• Particle collection occurs throughout the depth of the filter
Cellulose fiber	Typically used in limited/qualitative applications in air quality sampling	• Inexpensive • Convenient extraction of particulates	• Highly moisture sensitive • Limited temperature range • Low particle collection efficiency • Low chemical resistance
Borosilicate glass fiber	Wide scope in air quality sampling. Used without organic binder	• Temperature resistance to $\approx 500^{\circ}\text{C}$ • Chemically resistant to some extent	• Sulfate artifact formation due to alkalinity of fibers • Water vapor uptake can occur and must be equilibrated appropriately
Teflon coated glass fiber	Wide range of air sampling applications: emissions analysis, gravimetric analysis, biological analysis, and mutagenic analysis	• Low moisture uptake • Minimizes chemical transformation artifacts	• Artifact nitrate
Quartz fiber	Air sampling for chemical analysis of particulates: ion chromatography, atomic absorption, carbon analysis, PAH analysis, etc.	• Low moisture uptake • Stable to temperatures up to 800°C • Low trace contaminant levels • Can be baked to remove trace organics prior to sampling • Low artifact formation	• Friable • Artifact nitrate formation has been observed

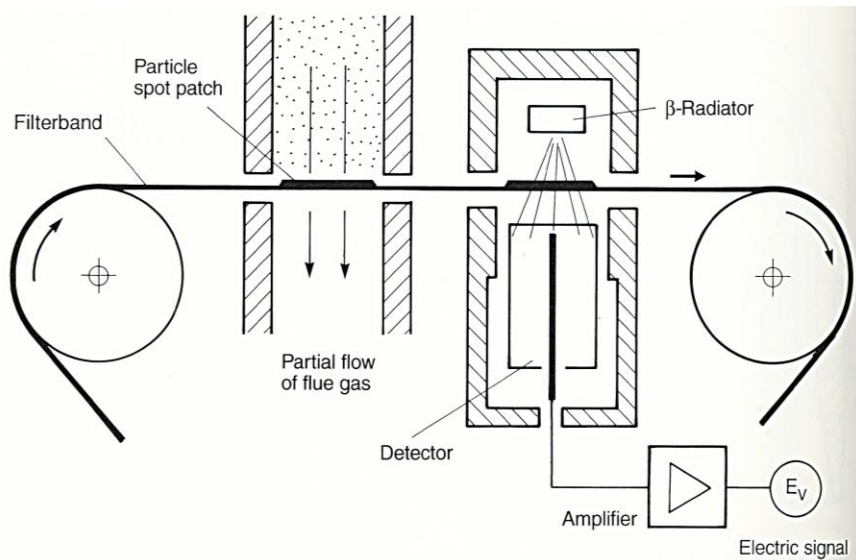
TABLE 7-7 Compilation of Common Applications, Advantages, and Disadvantages for Various Types of Membrane Sampling Filters

Filter Type	Typical Applications	Advantages	Limitations
Membrane filters (general) (apply to all below)	Used in air sampling for surface analytical techniques, submicrometer particle collection	• High collection efficiency • High mechanical strength	• High pressure drop • Low particulate loading capacity, rapid clogging • Limited temperature range, typically • Susceptible to water vapor uptake
Cellulose (mixed esters, nitrate, acetate, etc.) and PVC membranes	Sampling of metals, cotton dust, asbestos, etc., in NIOSH standard methods	• Inexpensive among membrane filters • Low chemical resistance	• Operating temperature limited to 75–130 °C • Electrostatic charge buildup observed in PVC membranes
Teflon membranes	Gravimetric analysis, neutron activities analysis XRF, XRD (see text)	• Inert to chemical transformations • Extremely low moisture sensitivity • Low trace/background concentrations • Chemical resistant	• Loss of nitrates observed • Temperature range limited to ≈ 150 °C for supported membranes and 260 °C for pure PTFE membranes

In-situ continuous methods

- Used when on-site continuous monitoring is needed
- In-situ Continuous methods for PM
 - Beta-attenuation monitor (BAM)
 - Filter tape method (with radiation attenuation mass detection)
 - Tapered Element Oscillating Microbalance (TEOM)
 - Change in mass collected changes frequency of oscillation
 - *Light scattering and absorption based*
 - *Nephelometer*
 - *Aethalometers*
 - *Optical particle counter (OPC)*

Beta-attenuation monitor



Met-one
E-BAM



Met-one
BAM-1022

Beta-attenuation monitor

$$\ln(I_0/I) = \int_0^d \mu dx$$

I_0 is the incident unattenuated beta counts, I is the attenuated beta counts through a substrate of mass density x (mg cm⁻²) and μ is mass absorption or attenuation coefficient (cm² mg⁻¹)

Mass absorption coefficient:

$$\mu \text{ (cm}^2/\text{g)} = 0.016 (Z/A)^{4/3} E_{\text{max}}^{-1.37}$$

Z/A: atomic number to mass ratio

TABLE 12-2 Effect of Atomic Number Dependence on the Measured Mass of Several Compounds

Compound	Z/A	μ (cm ² /mg) ^a	μ (cm ² /mg) ^b
(NH ₄) ₂ SO ₄	0.530	0.153	0.166
NH ₄ HSO ₄	0.521	0.152	0.163
CaSO ₄ · 2H ₂ O	0.511	0.152	0.159
SiO ₂	0.499	0.154	0.154
CaCO ₃	0.500	0.154	0.154
Carbon	0.500	0.154	0.154
Fe ₂ O ₃	0.476	0.163	0.144
NaCl	0.478	0.172	0.145
PbSO ₄	0.429	0.193	0.126
PbCl ₂	0.417	0.204	0.121
PbBrCl	0.415	0.206	0.120

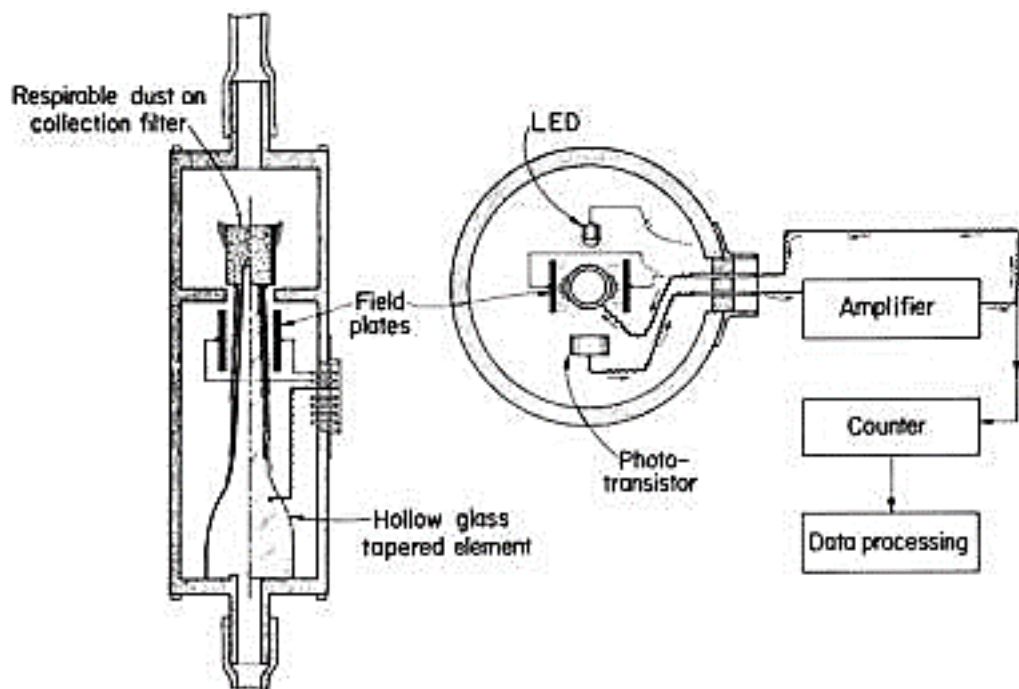
^aData from Jaklevic et al. (1981).

^bZ/A dependence calculated from Equation 12-2. Values normalized to 0.154 for carbon to account for instrumental differences.

Beta-attenuation monitor

- Potential biases
 - Atomic number dependence of the mass absorption coefficient.
 - Fluctuations in atmospheric pressure, changes in laboratory temperature and relative humidity.
 - Positive sampling artifacts (absorption and adsorption of gaseous compounds on sampled particles and filter media) and from negative sampling artifacts (the loss of semi-volatile material during sampling).

Tapered Element Oscillating Microbalance



Thermo Scientific™ 1405 TEOM™

TEOM principle

The equation which describes the tapered-element oscillating microbalance (derived from simple harmonic oscillator) is

$$\Delta m = K_0 (1 / f_b^2 - 1 / f_a^2)$$

where,

Δm is the mass of the collected sample;

f_a is the frequency of the oscillating element before sample collection;

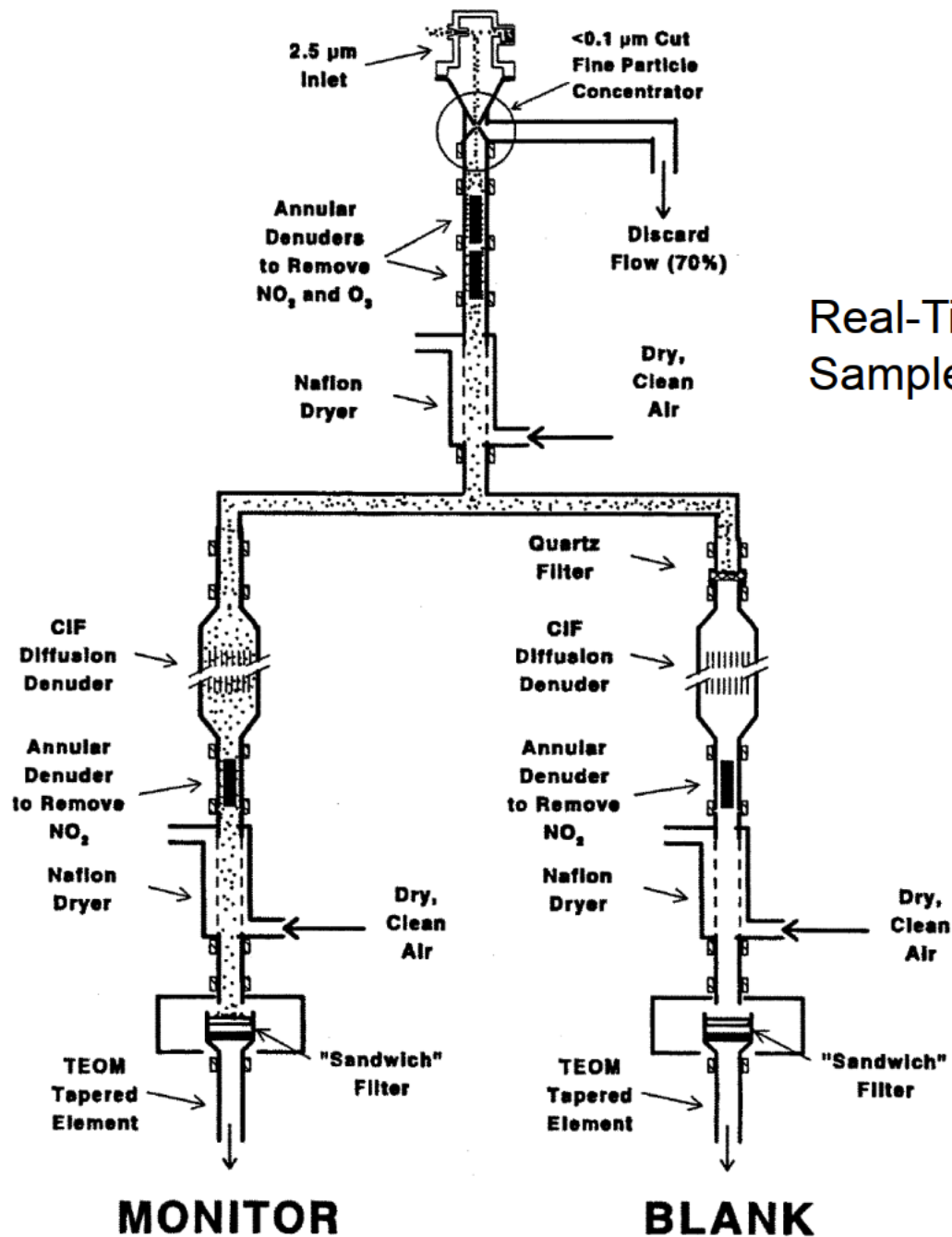
f_b is the frequency of the oscillating element after sample collection;

K_0 is a constant (spring constant) unique to each tapered element

Piezoelectric effects

(Quartz crystal microbalance)

- Piezoelectric: when a quartz crystal is mechanically stressed, electrical charges of opposite polarity appear on opposite surfaces of the crystal.
- The reverse is also true: when crystalline quartz is in an electrical field, it becomes mechanically stressed in a direction depending upon the polarity of the field.
- The reverse effect can result in the oscillation of the crystalline quartz with a certain frequency



Real-Time Ambient Mass Sampler (RAMS)

Eatough et al. (2001)

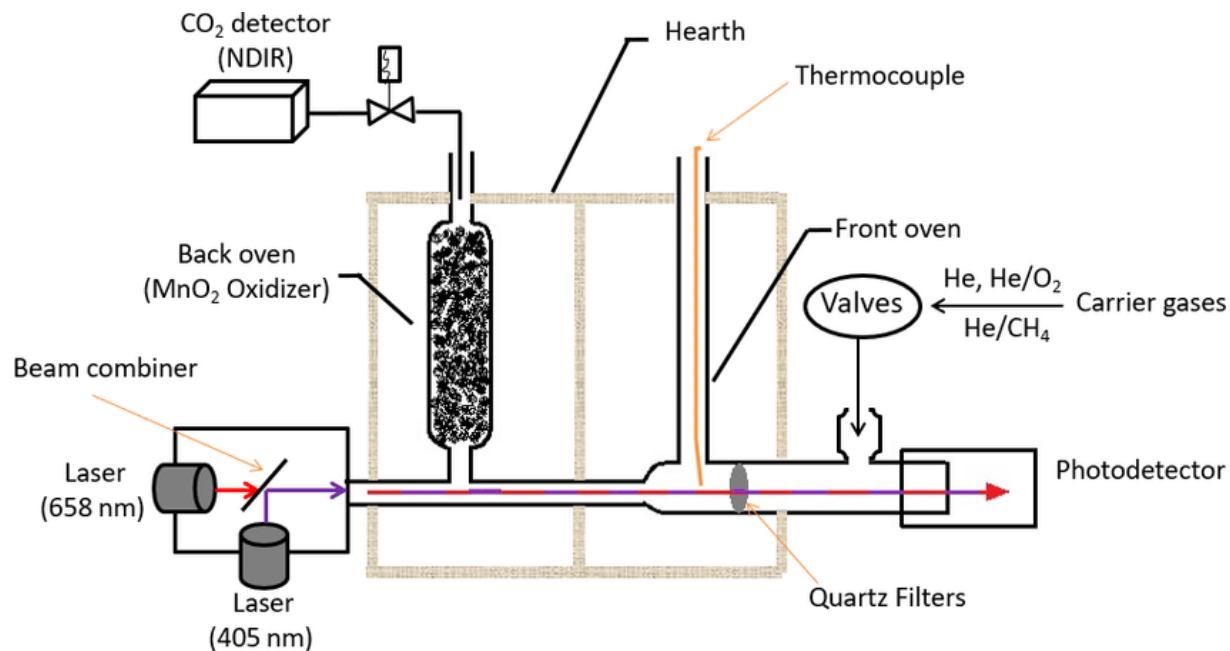
TEOM versions

- Differential Tapered-Element Oscillating Microbalance
 - A size-selective inlet followed by a diffusion dryer, an ESP, and the TEOM mass sensor operated at reduced filter temperature (e.g., 30°C).
 - The ESP is alternately switched on and off (for equal time periods, e.g., 5 min).
 - Correction of volatilization of previously collected particle constituents and adsorption of gas species.
- TEOM Equipped with a Filter Dynamics Measurement System
 - A cold filter maintained at 4°C instead of an electrostatic precipitator to remove particles.
 - Alternate every 6 min between ambient air sampling (base flow) and purging the sample filter with particle-free air (reference flow).

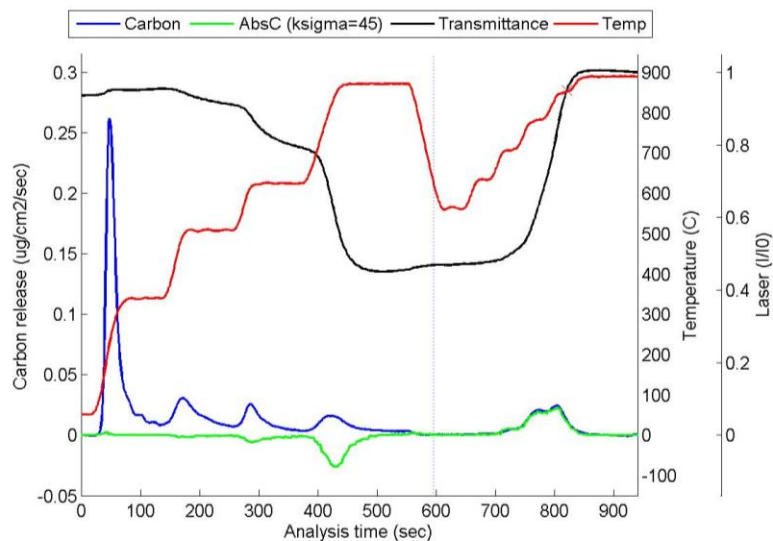
TEOM Potential biases

- Adherence Effects
- Overloading
- Electrostatic Effects
- Volatilization Losses

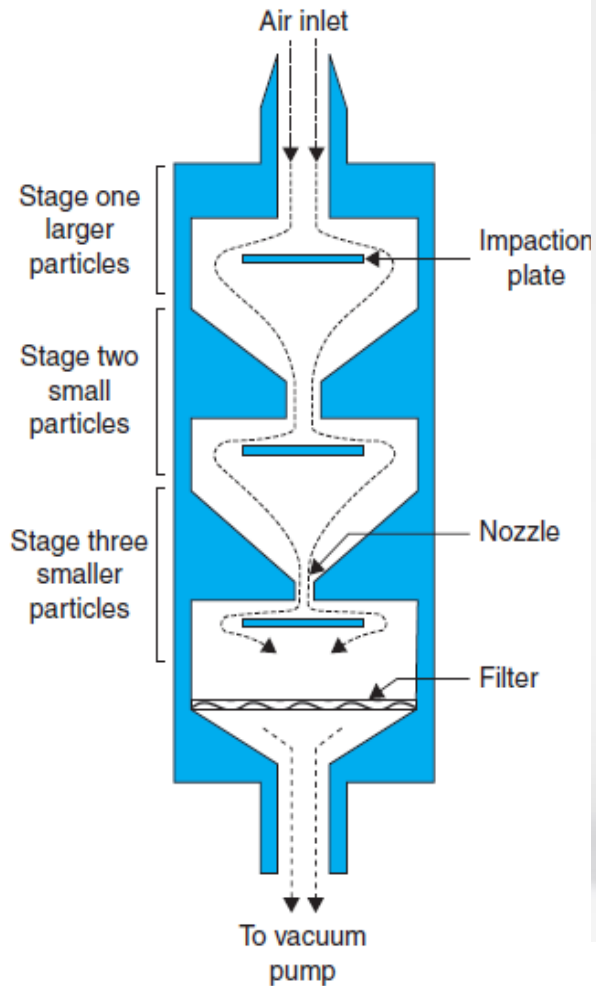
OC/EC analyzer



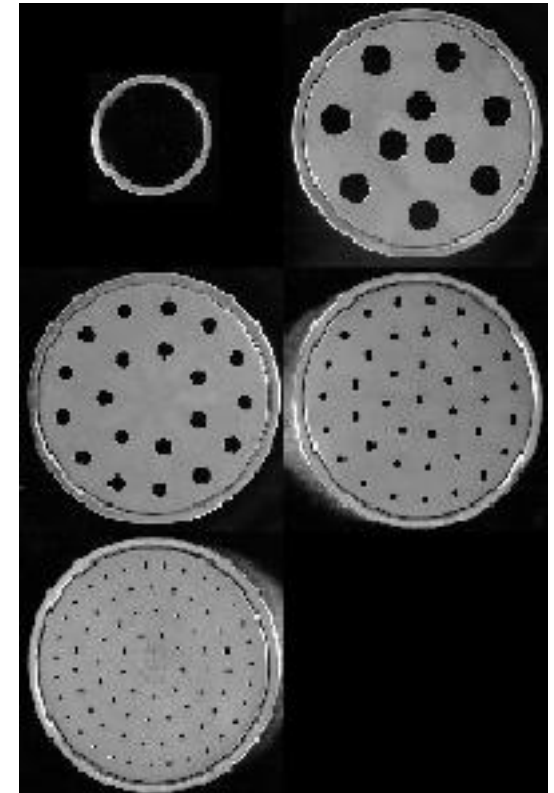
He: obtain OC
O₂: obtain EC



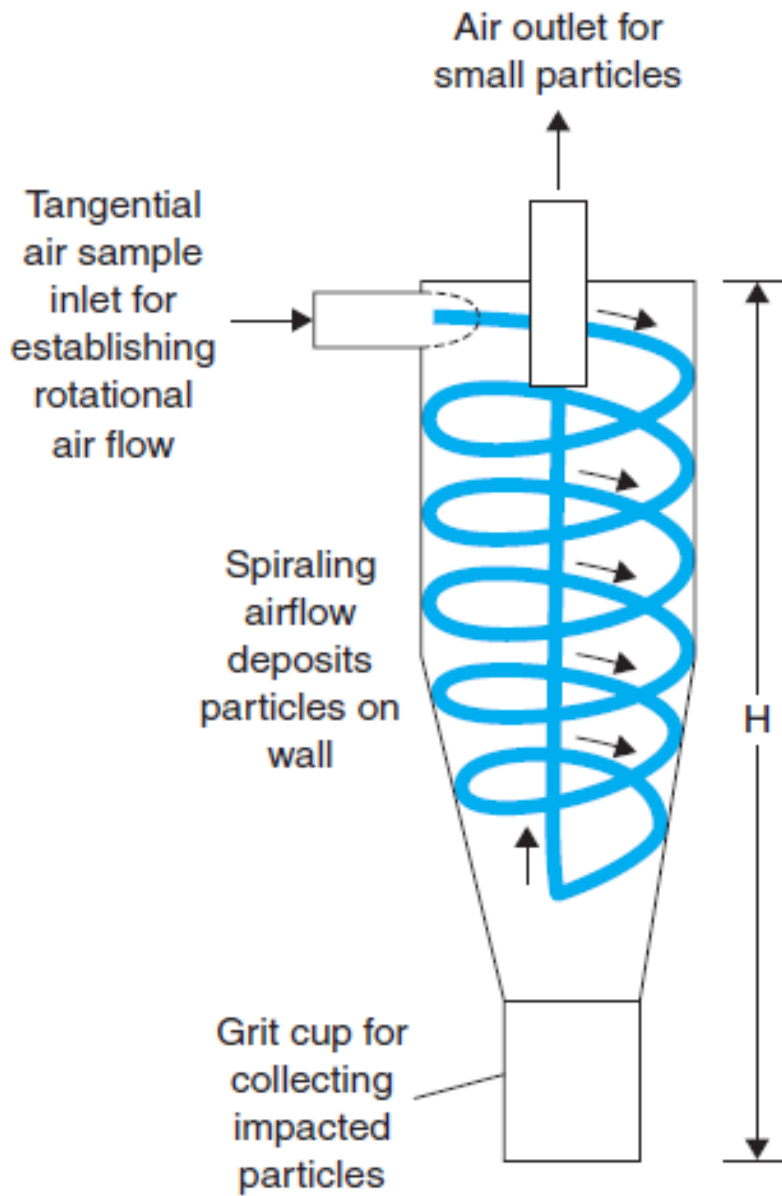
Cascade impactor



(A)



Cyclone pre-collector



Source: The University of California Air Pollution Health Effects Laboratory, with kind permission.